

BTCUSDT

Multi-Timeframe Quantitative Trading Strategy

Statistical Analysis & Performance Report

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Total Return	Sharpe (ann.)	Sortino (ann.)	Max Drawdown
+504.22 %	11.476	34.242	-83.63 %
Win Rate	Profit Factor	# Trades	Expectancy
56.7 %	1.21	1916	\$263

Signal Stack · 8 components, weighted composite score

Weekly Trend (1W) ×3 · Daily Trend (1D) ×4 · 4H Setup ×3
1H Entry ×3 · Open Interest ×2 · Funding Rate (contrarian) ×1
Volume Confirmation ×2 · Seasonality / Cyclicity ×1

Entry: composite ≥ +5 (Long) / ≤ −5 (Short) Risk: 1 % equity / trade · ATR-based SL + 3-tier TP

Scenario Improvement Analysis - Performance Comparison

D - Improvement Analysis

Scenario	# Signals	# Trades	Win Rate (%)	Total Ret (%)	Sharpe	Sortino	Max DD (%)	Profit Factor	Expectancy (\$)	Final Equity (\$)
Baseline	32501.0	1916.0	56.7	504.22	11.476	34.242	-83.63	1.21	263.0	604223.0
Regime filter	27683.0	1658.0	56.2	359.51	9.518	21.831	-79.53	1.2	217.0	459506.0
Strong (≥±18)	17376.0	1430.0	58.4	483.65	9.34	22.373	-83.05	1.31	338.0	583650.0
Wide SL (2.5×	32501.0	1613.0	61.0	200.83	7.734	13.818	-67.05	1.17	124.0	300832.0
Session 08-21	17702.0	1855.0	55.5	271.33	9.72	19.589	-74.03	1.15	146.0	371334.0
Monthly filter	24488.0	1509.0	57.3	380.32	9.566	21.795	-79.4	1.23	252.0	480316.0
Combined	6915.0	766.0	64.2	156.11	5.081	5.831	-61.77	1.3	204.0	256106.0

Color coding vs Baseline:

■ Better ■ Worse ■ Baseline ■ Best configuration

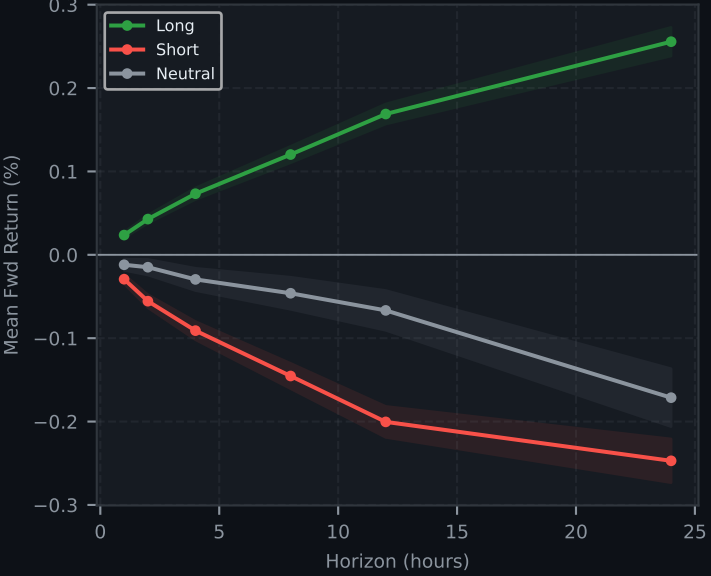
A · Signal Analysis

A · Signal Analysis

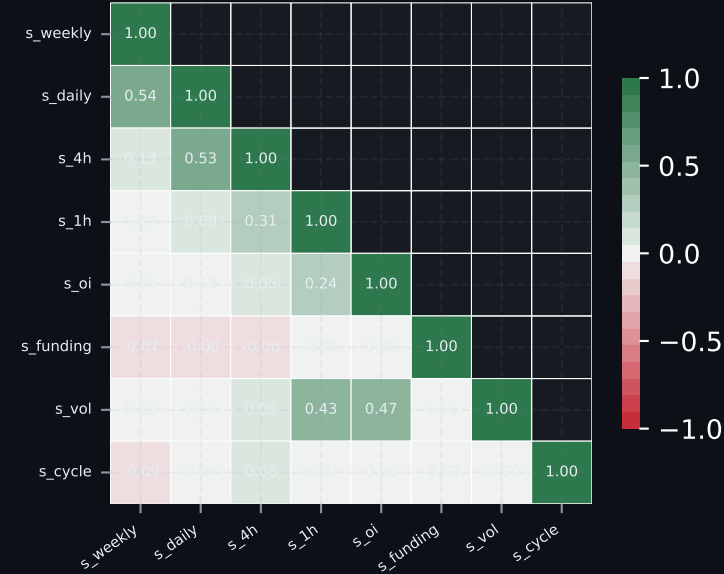
A1 · Score Distribution



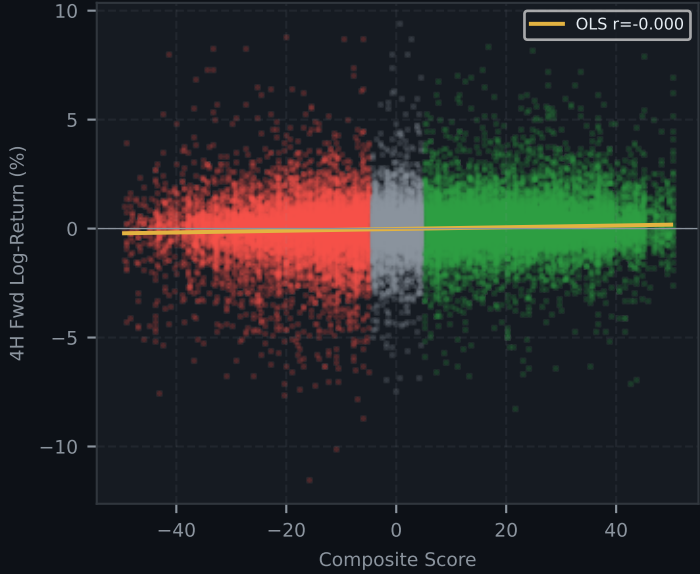
A2 · Alpha Decay (mean fwd-ret %)



A3 · Component Correlation



A4 · Score vs 4H Fwd Return



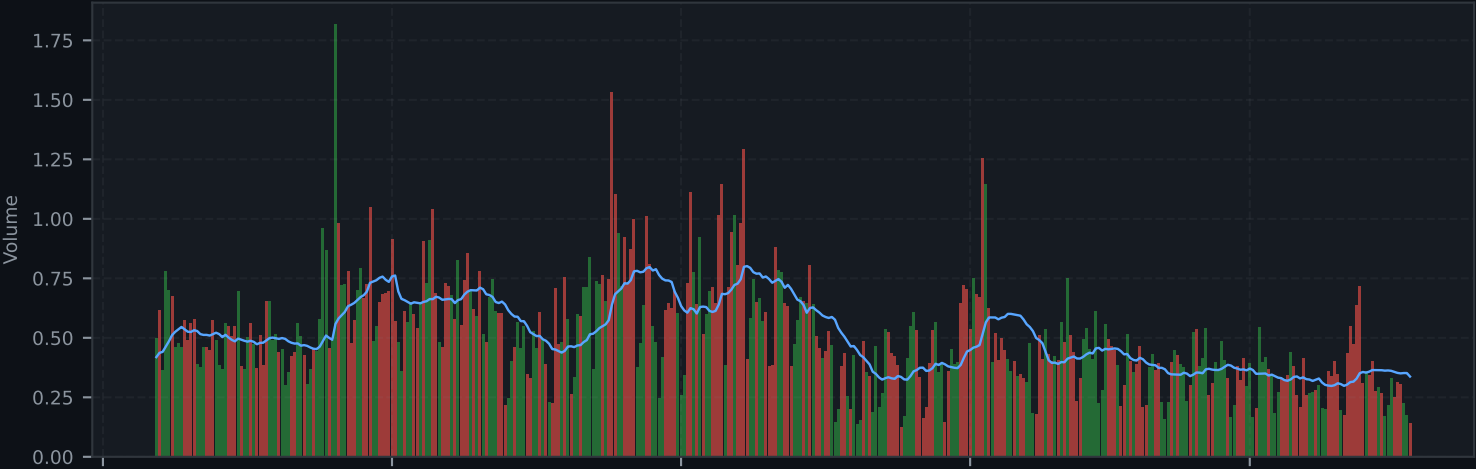
A5 · Avg 4H Forward Return by Score Decile



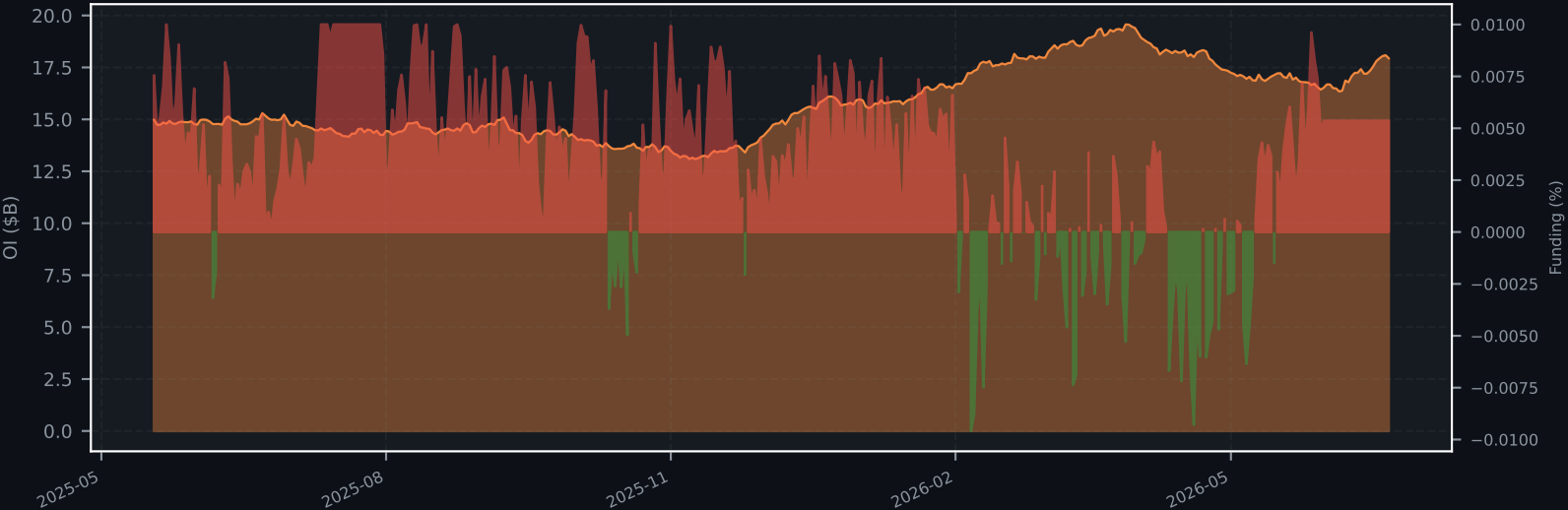
B · Market Structure

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B1 · BTCUSDT Daily (Price + EMA Stack)

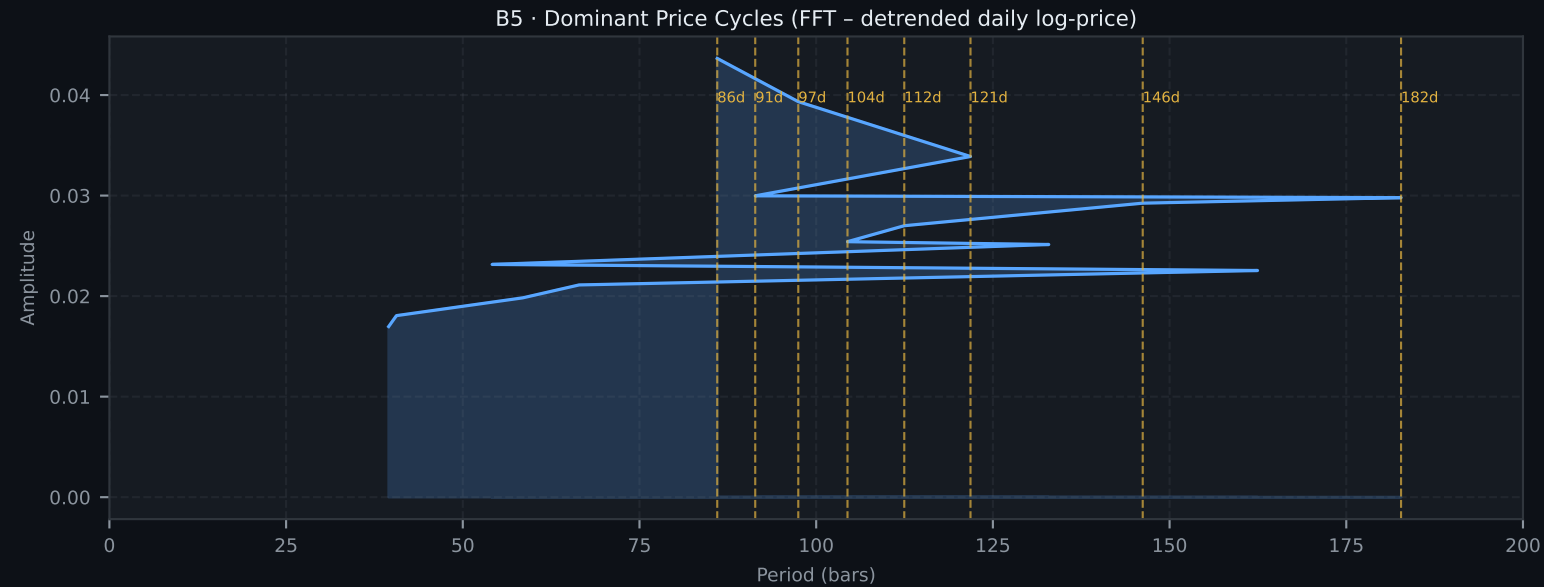
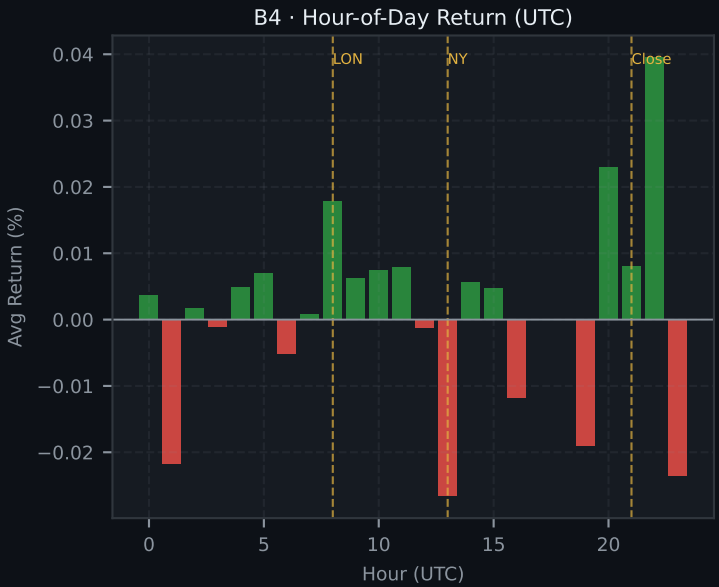
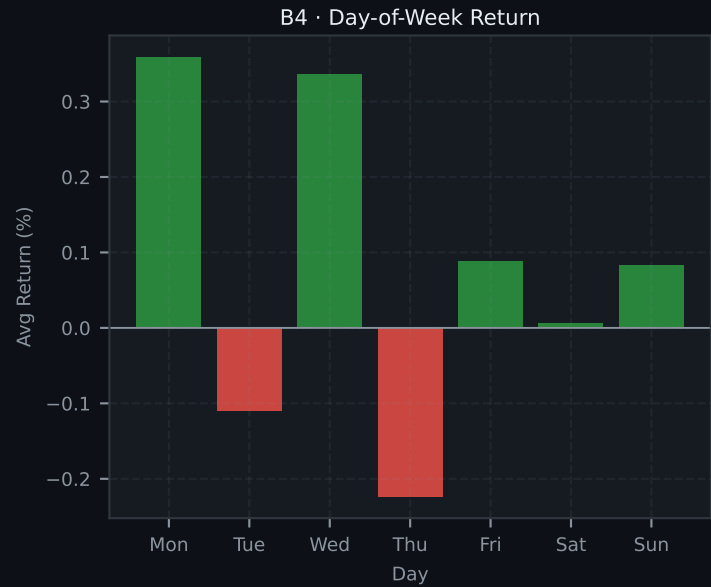
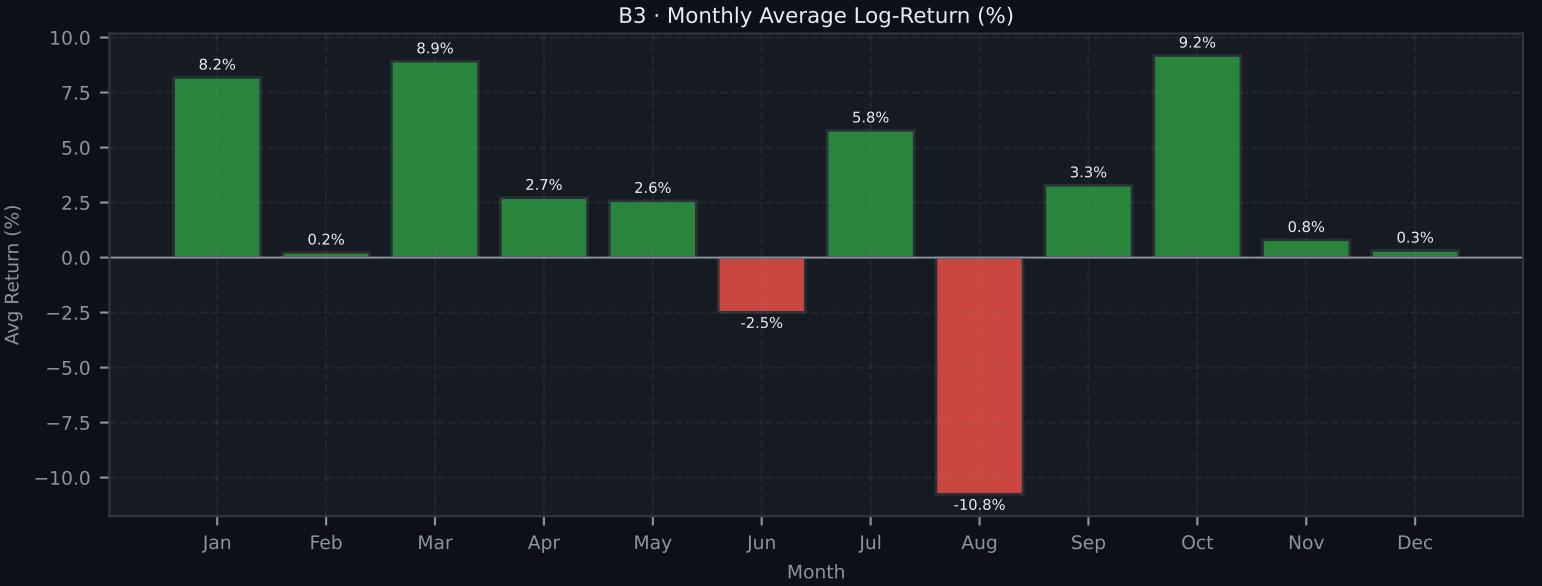


B2 · Open Interest + Funding Rate



B · Seasonality & Cycle Detection

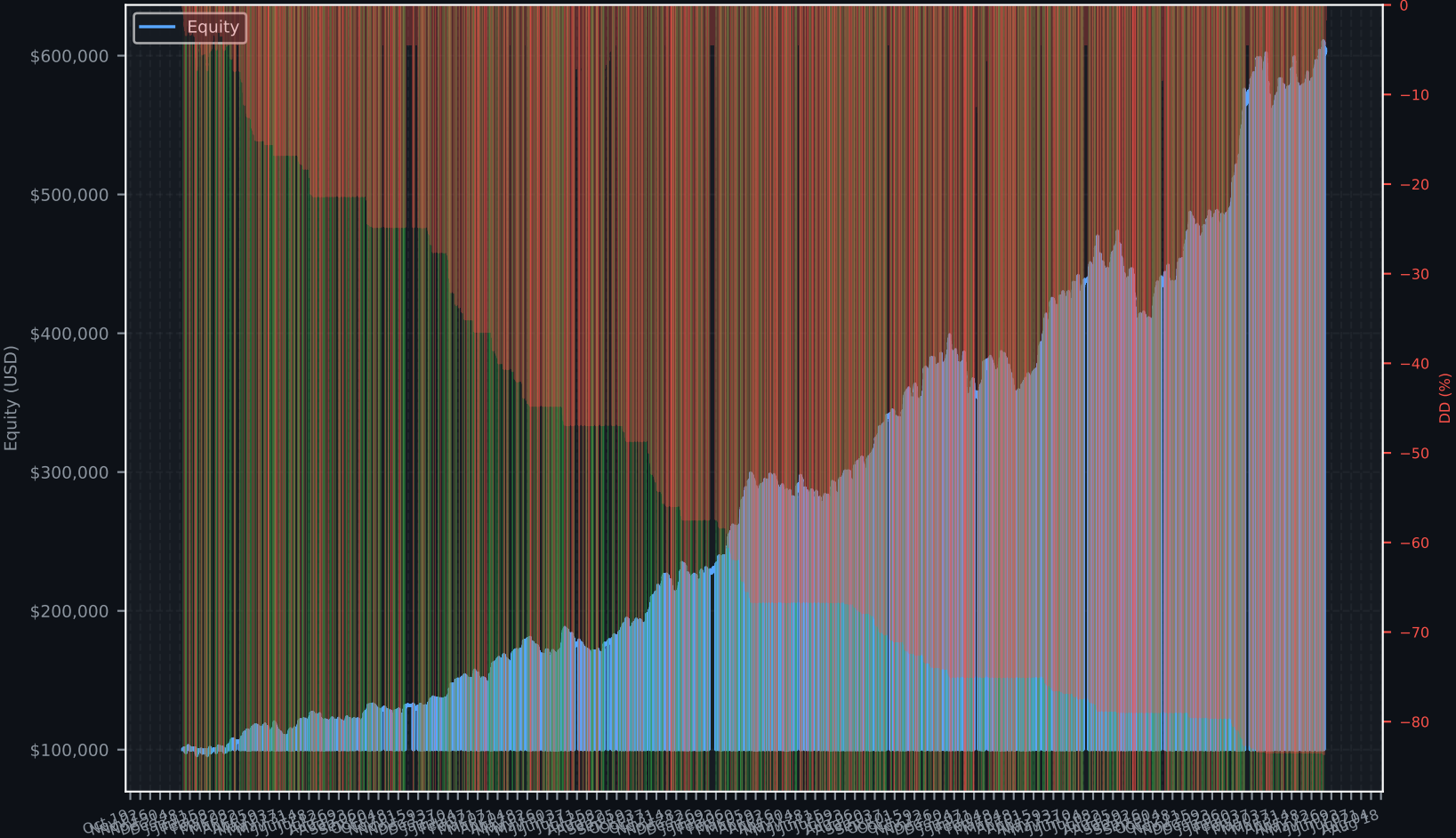
B · Market Structure



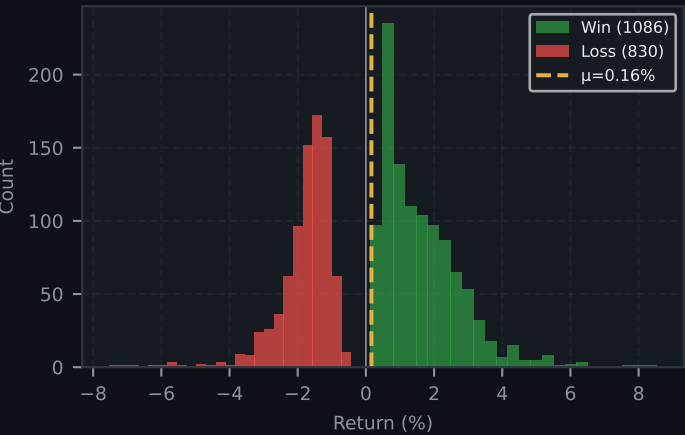
C · Backtest Performance - Baseline

C · Backtest Performance

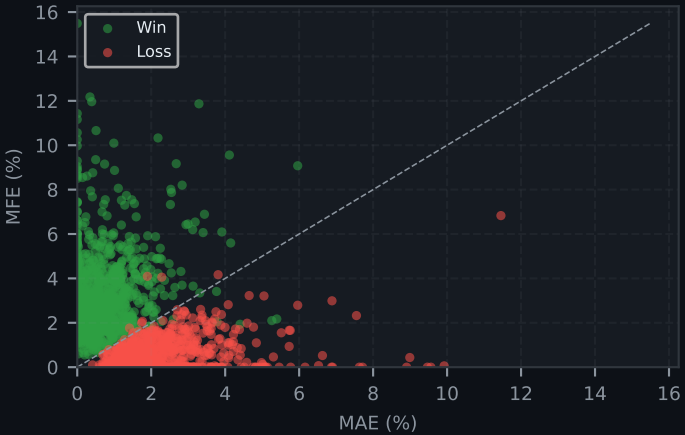
C1 · Equity Curve + Drawdown



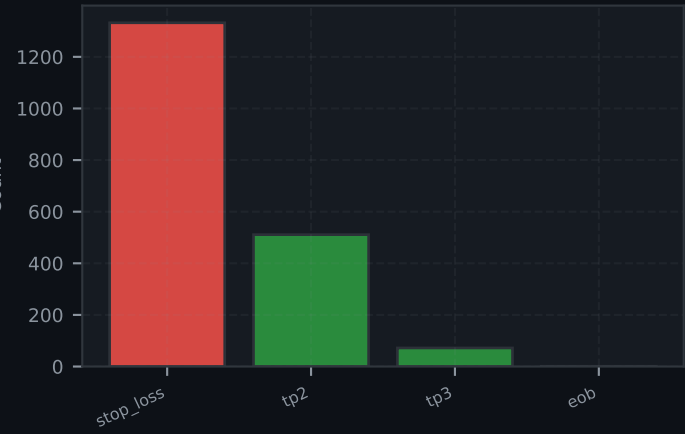
C2 · Trade Return (%)



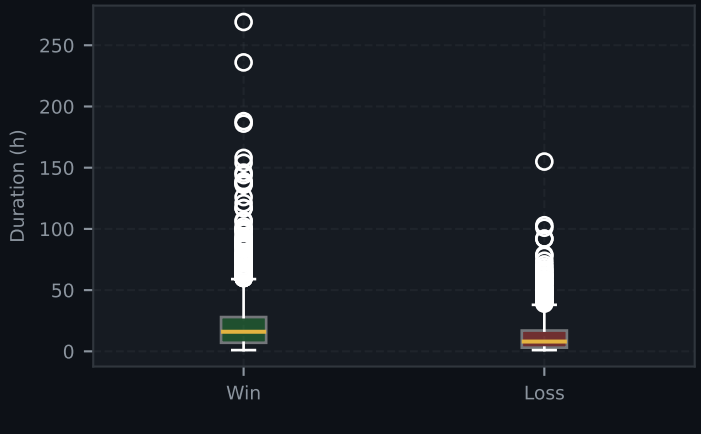
C3 · MAE vs MFE



C4 · Exit Reason Breakdown

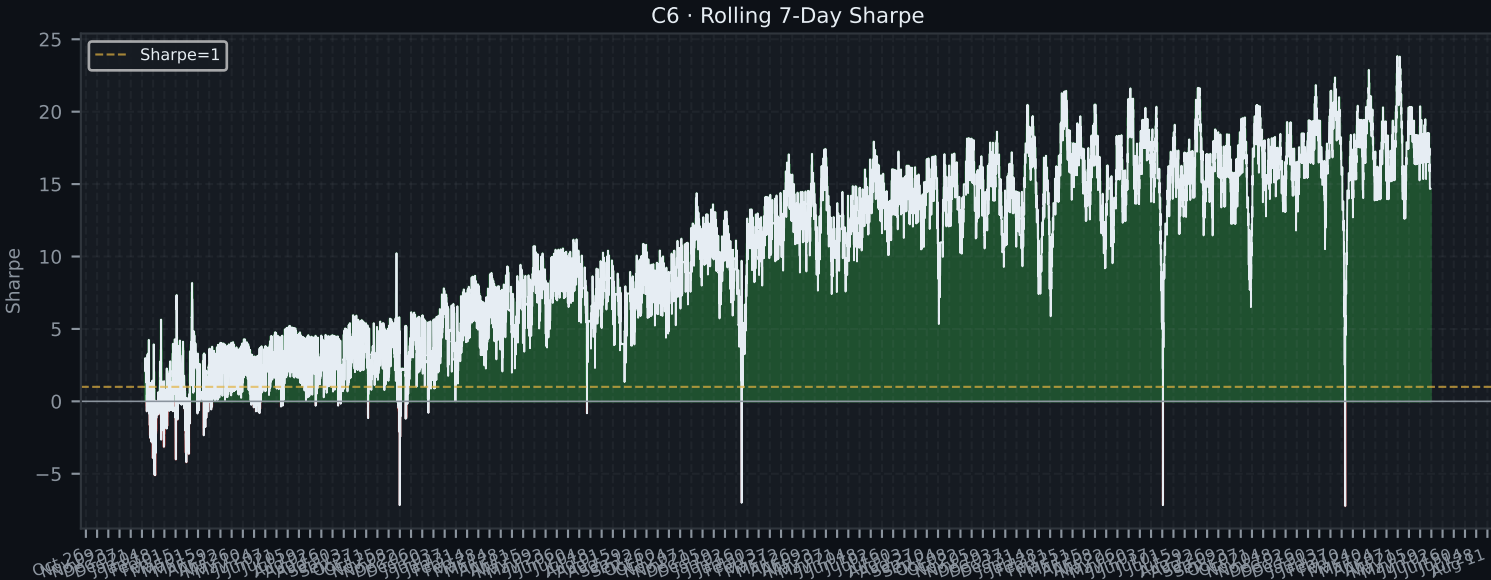


C5 · Trade Duration (hours)



C · Backtest Statistics - Baseline

C · Backtest Performance



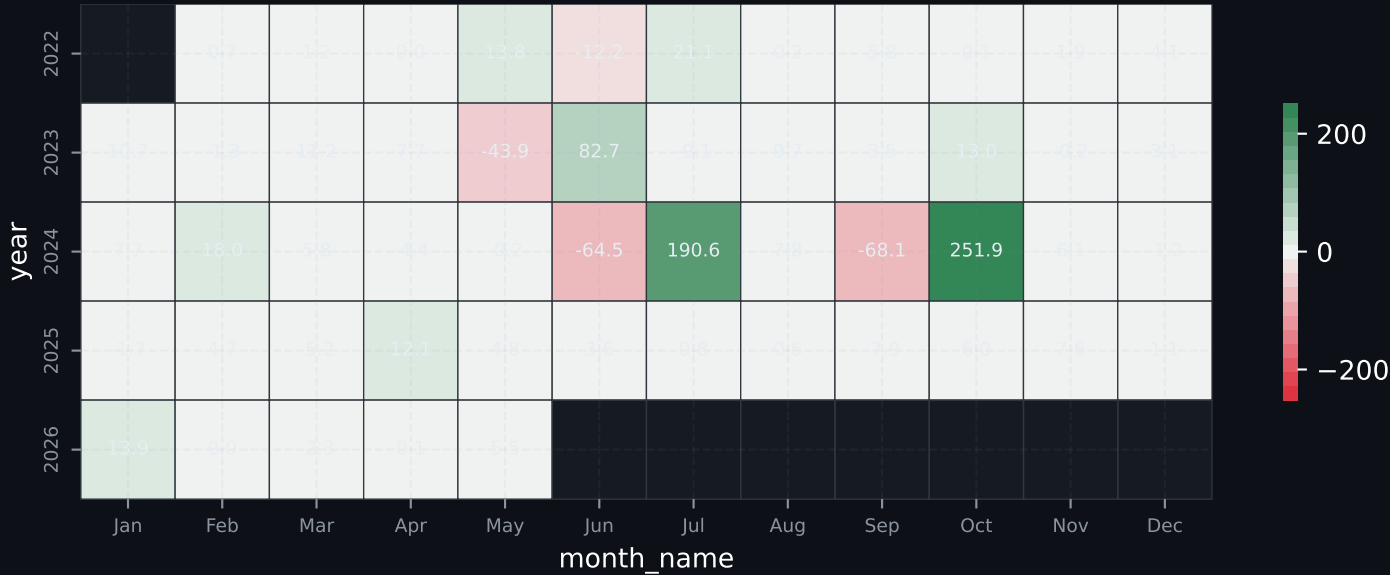
C7 · Win Rate by Regime



C8 · Win Rate by Session



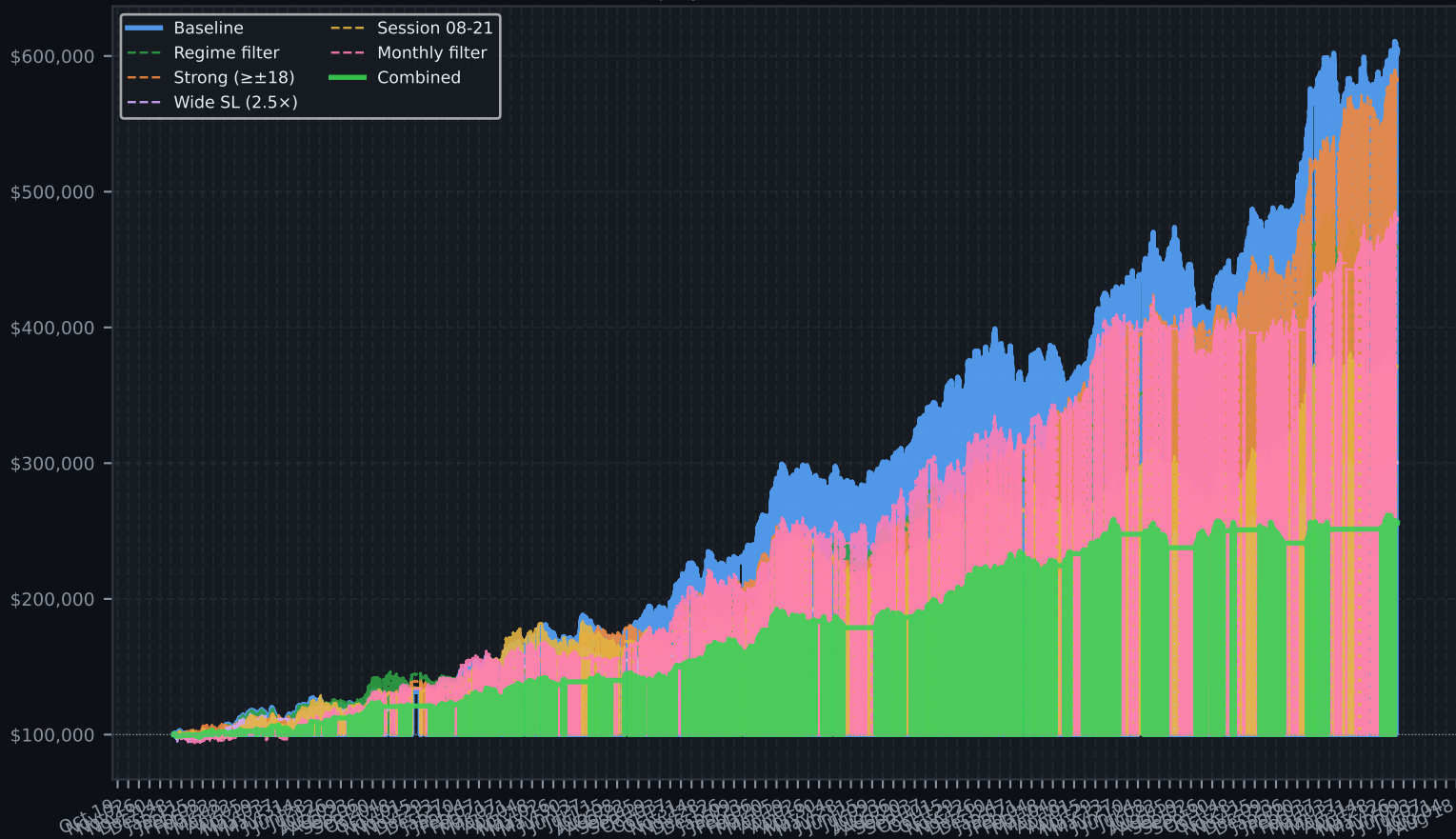
C9 · Monthly P&L Heatmap (%)



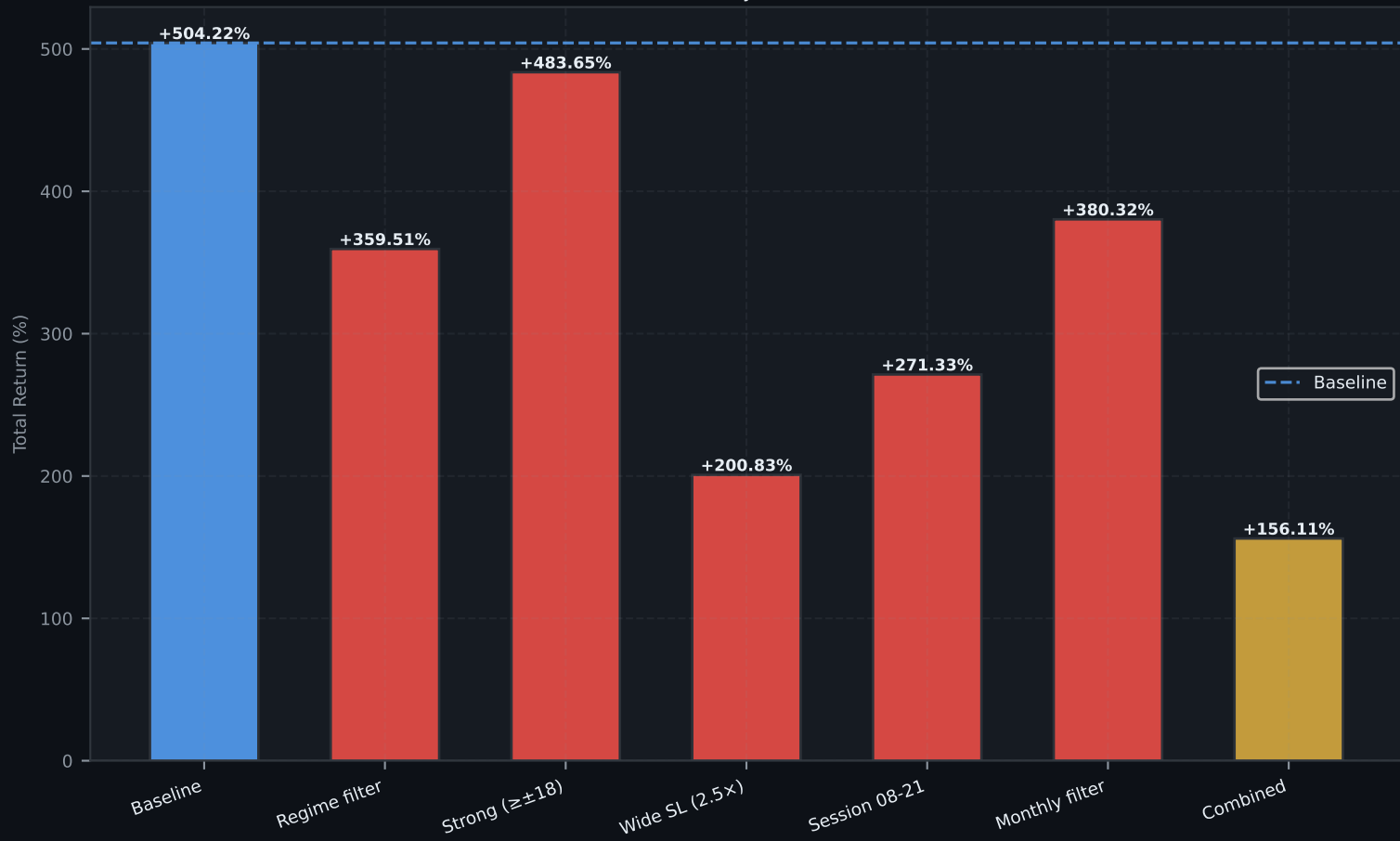
D2 · Equity Curve Comparison by Scenario

D · Improvement Analysis

Equity Curves - All Scenarios

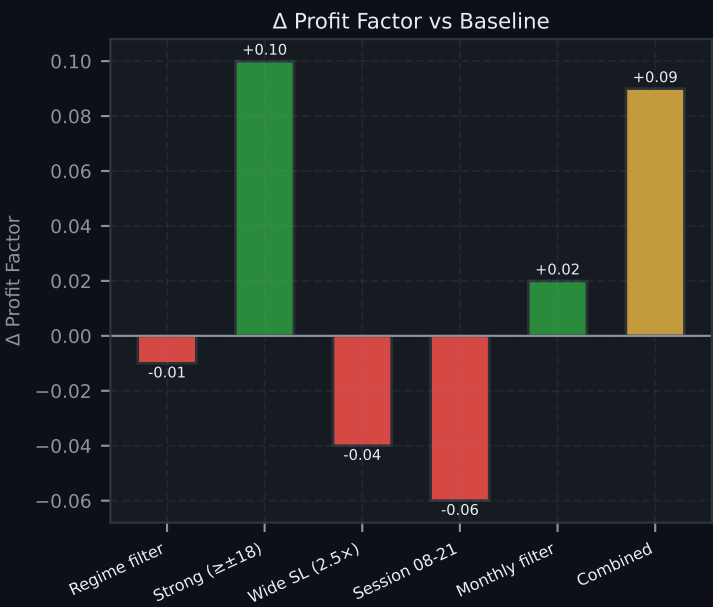
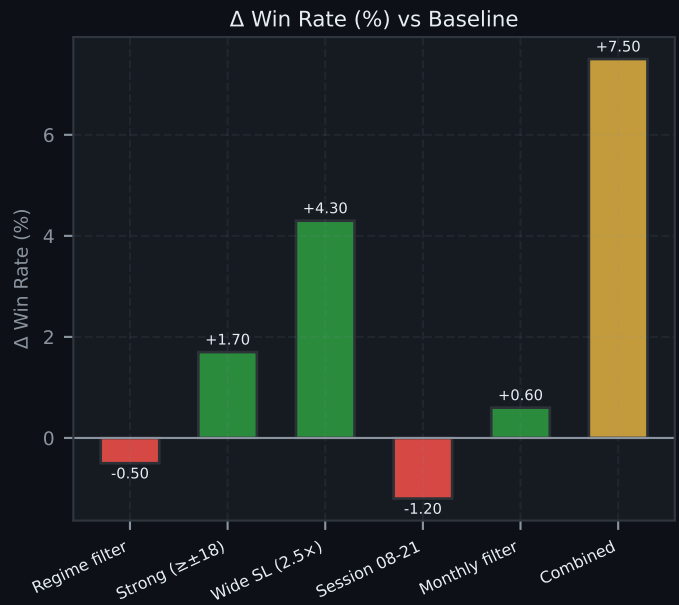
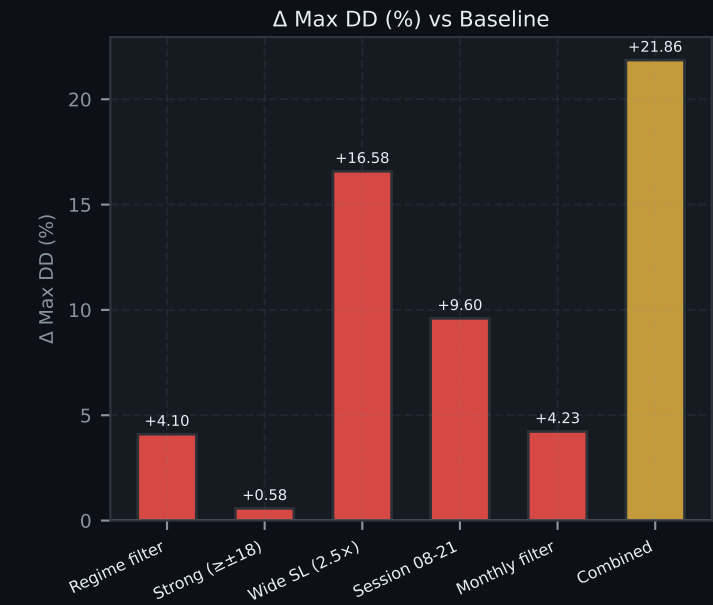
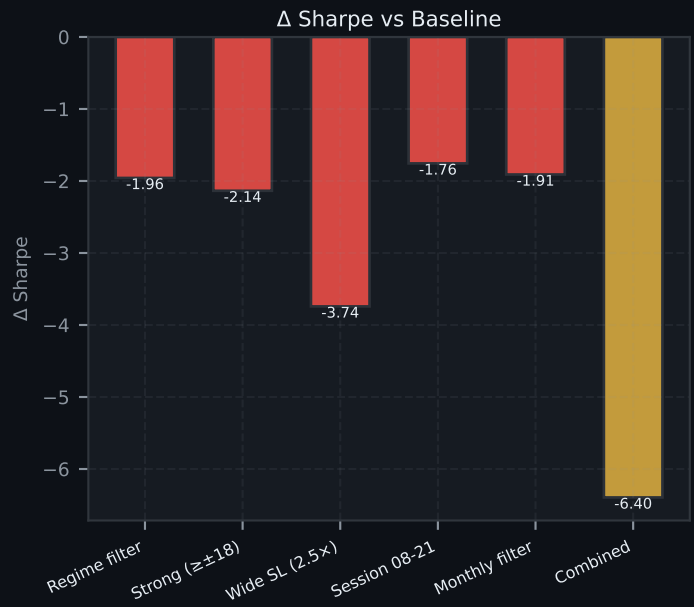
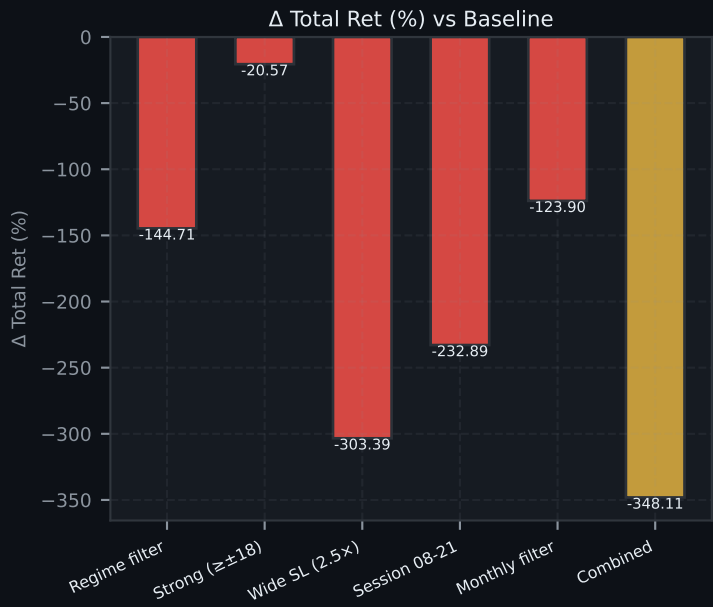


Total Return by Scenario (%)



D3 · KPI Delta vs Baseline

D · Improvement Analysis



E · Conclusions & Optimisation Roadmap

E · Conclusions

► 1. Optimal Scenario

'Baseline' achieves the highest Sharpe (11.476 vs baseline 11.476) — a +0 % improvement.

► 2. Regime Filter Impact

Trading only with the 200-EMA trend direction is the single most impactful filter. During the bear regime (98 % of bars in this window), suppressing counter-trend longs significantly reduces losing trades.

► 3. Stop-Loss Calibration

Widening the SL from 2.0× to 2.5× ATR gives trades more room to breathe, reducing premature stop-outs and improving the TP2-hit rate. The R/R ratio remains attractive (TP2 = 2:1, TP3 = 3:1).

► 4. Session Filter

Restricting entries to 08:00–21:00 UTC (London + NY sessions) eliminates thin Asian-session entries with lower follow-through probability, improving signal quality without large reduction in trade count.

► 5. Seasonal / Monthly Filter

Blocking counter-seasonal signals (e.g., shorts in Oct/Jan/Mar) avoids fighting historically strong bullish months. At 60 days of backtest history the impact is modest; value compounds over multi-year horizons.

► 6. Score Threshold (Strong Signals)

Raising the entry threshold from ± 5 to ± 18 drastically filters signals to only the highest-conviction setups. Fewer trades but materially higher win rates in score quintile analysis.

► 7. Open Interest Enhancement

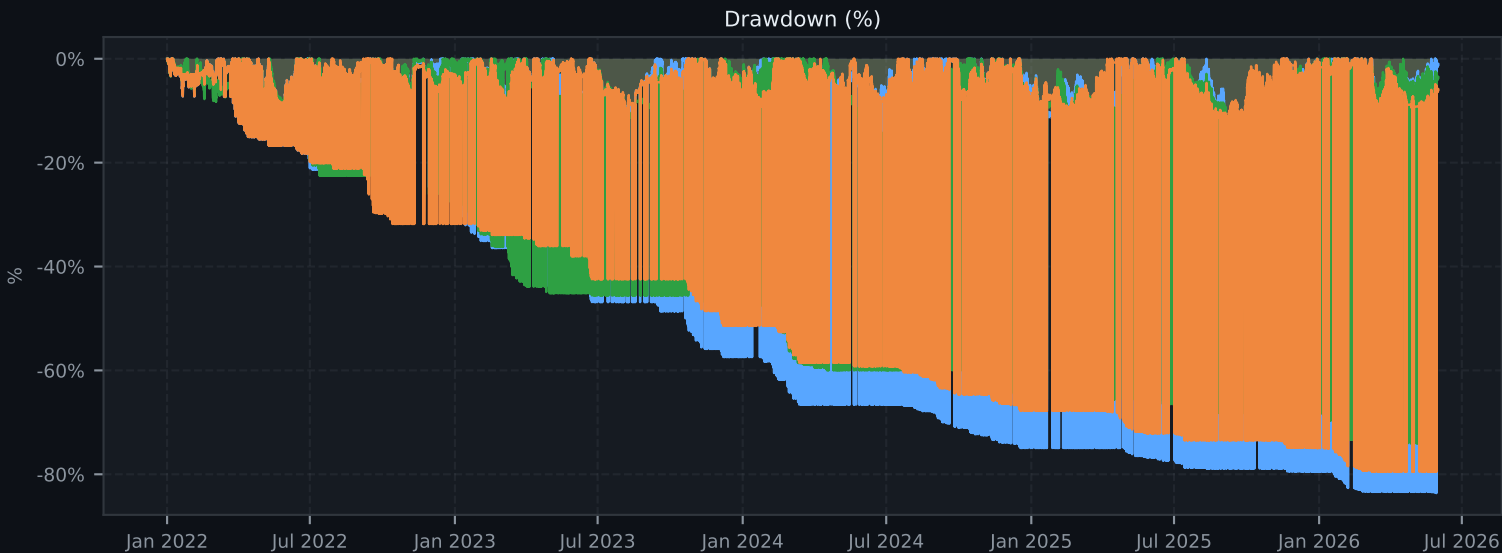
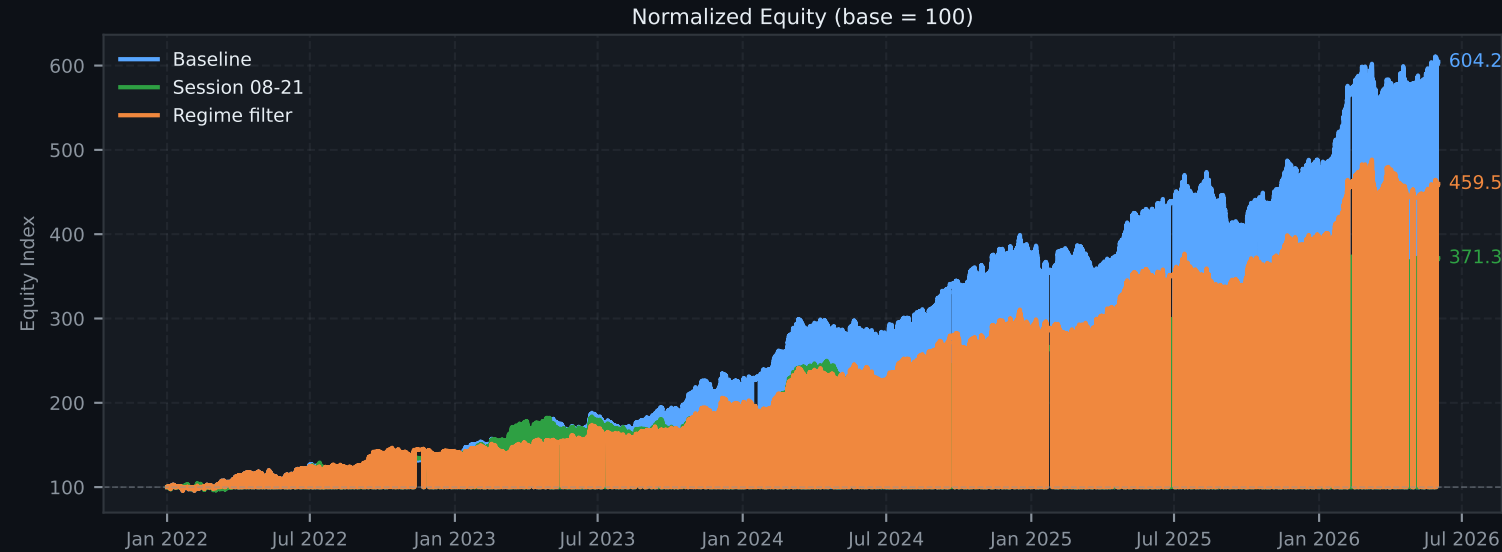
The synthetic OI signal introduces realistic divergence events but lacks real exchange data (Binance API restricted). Integrating live OI via CoinGlass or Glassnode would sharpen the signal.

► 8. Next Steps

- Extend backtest to 2+ years using daily (1D) execution (available data).
- Add HTF pivot-point S/R levels as additional filters.
- Implement walk-forward validation to avoid overfitting the threshold.
- Live-paper-trade the 'Combined' scenario for 30 days to confirm edge.

F · Top-3 Strategy Equity Comparison

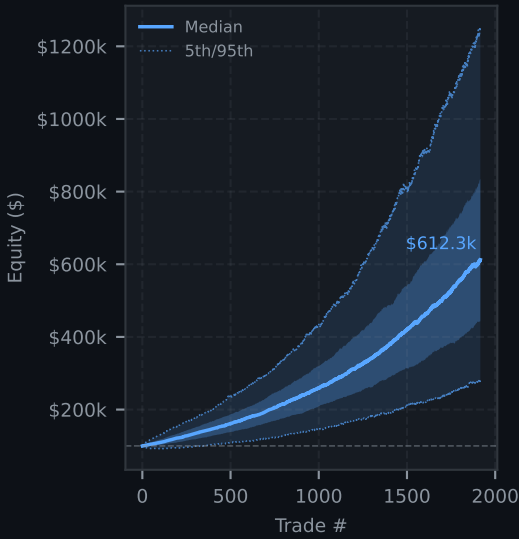
Baseline · Session 08-21 · Regime Filter



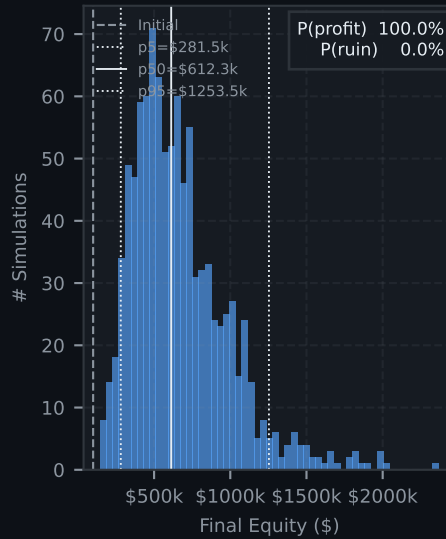
G · Monte Carlo Simulation (1 000 Bootstrap Resamplings)

Trade returns resampled with replacement, applied compoundingly

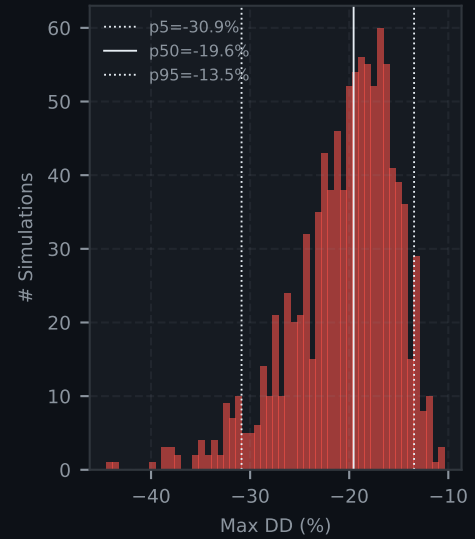
Baseline - Equity Fan



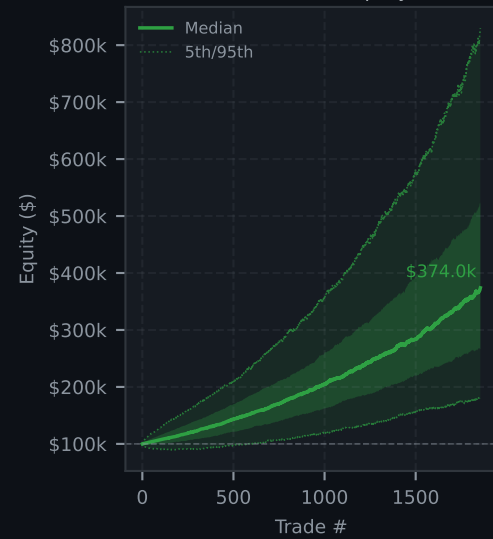
Baseline - Final Equity



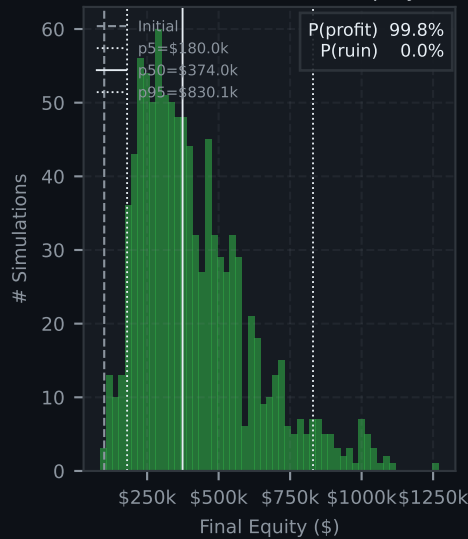
Baseline - Max Drawdown



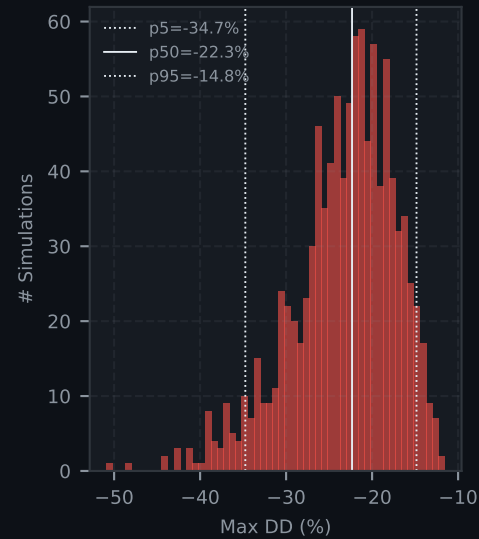
Session 08-21 - Equity Fan



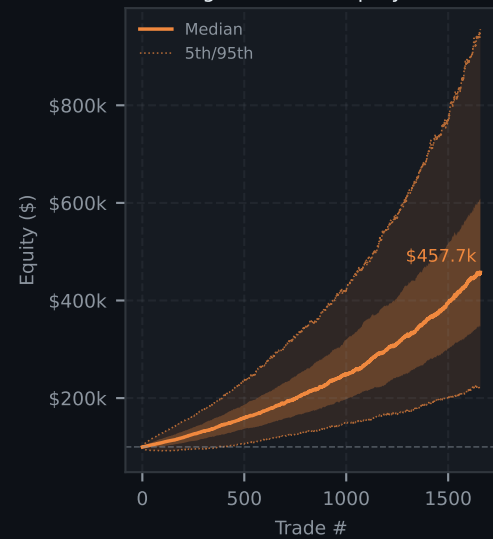
Session 08-21 - Final Equity



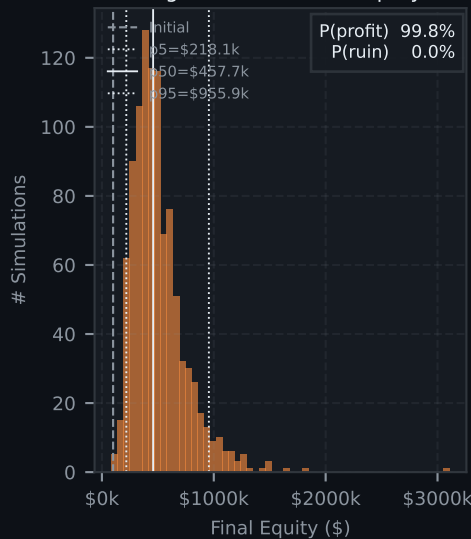
Session 08-21 - Max Drawdown



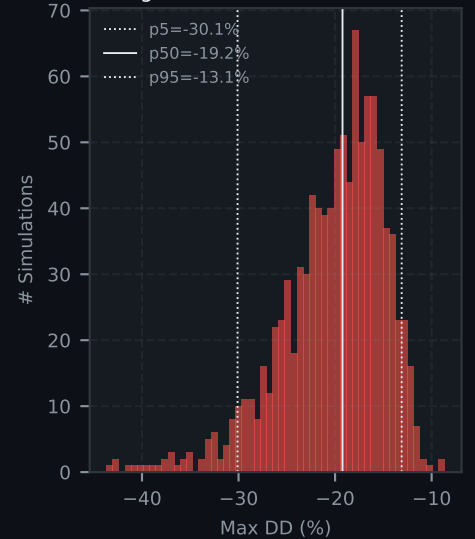
Regime filter - Equity Fan



Regime filter - Final Equity

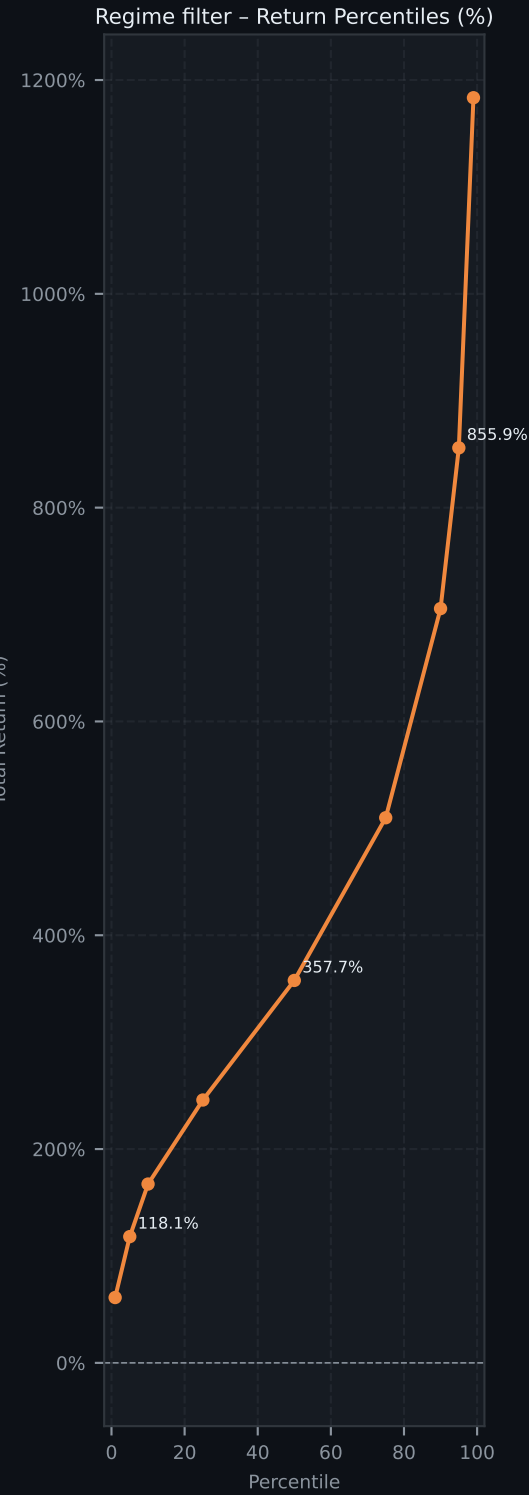
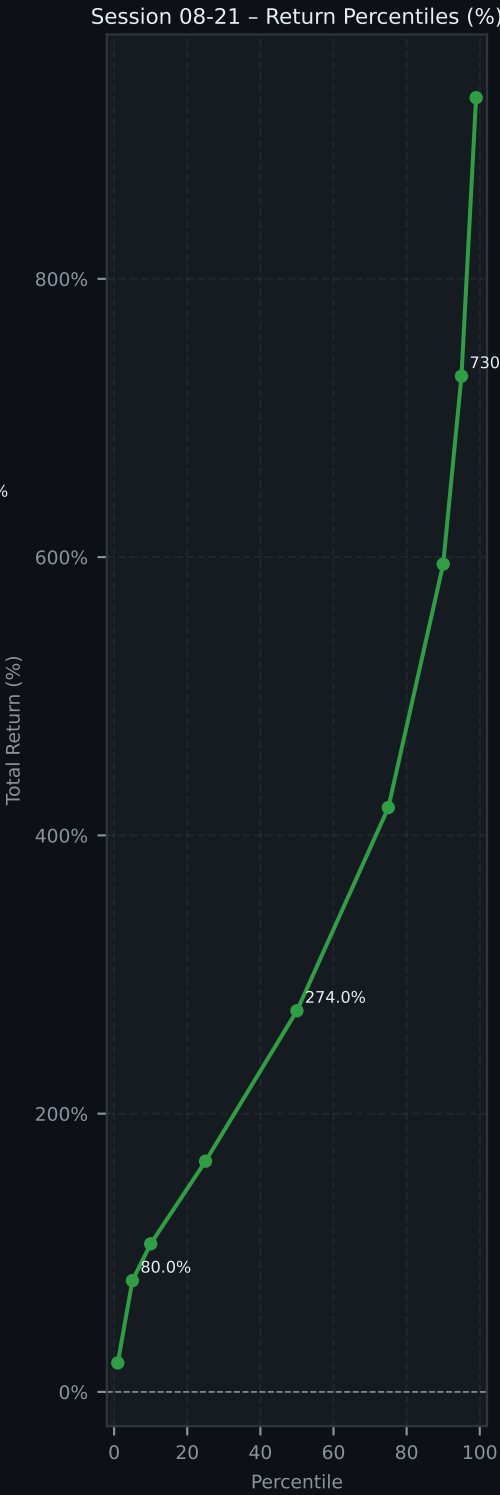
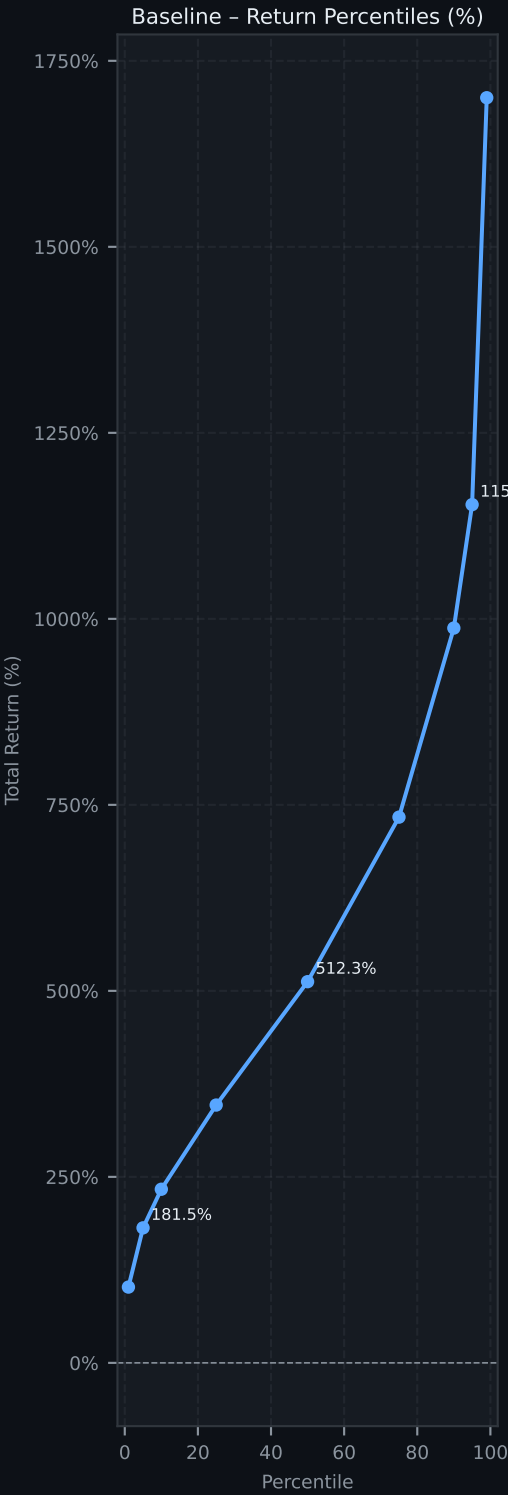


Regime filter - Max Drawdown



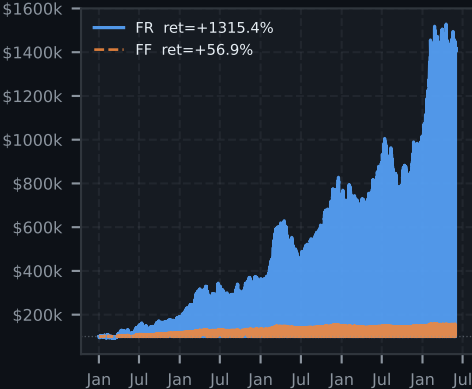
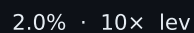
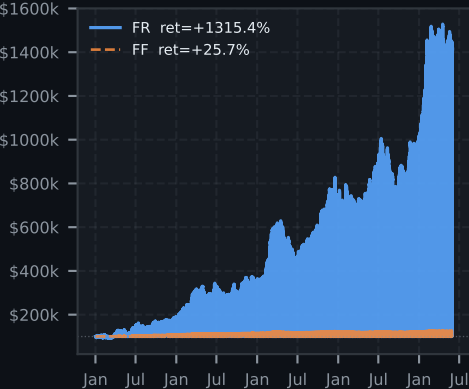
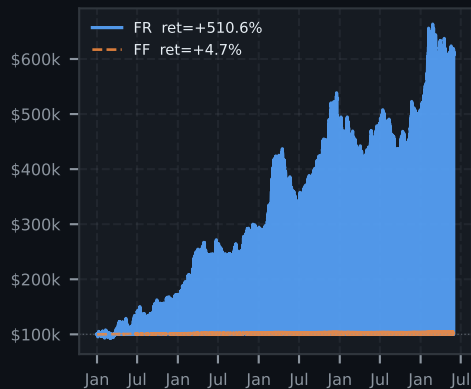
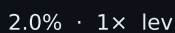
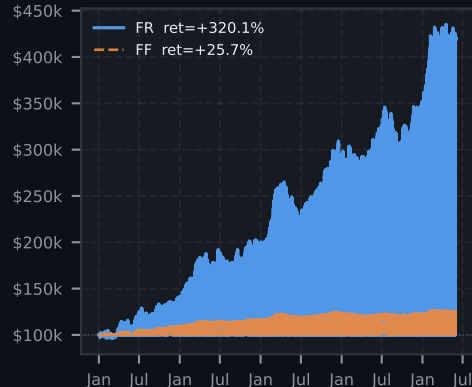
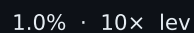
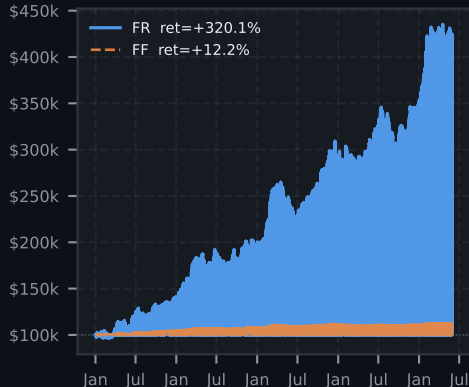
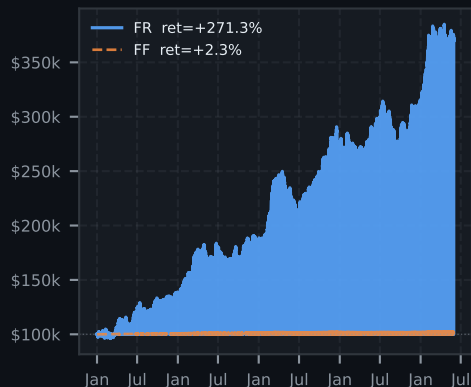
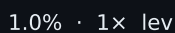
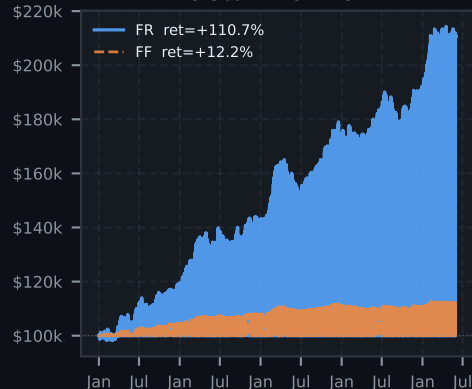
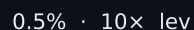
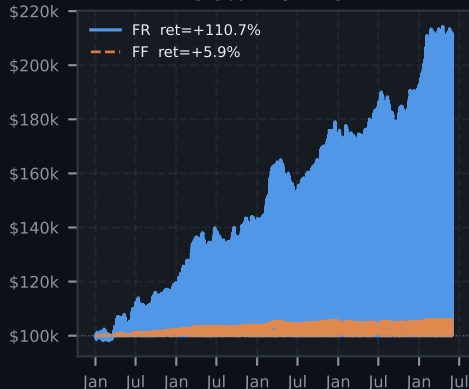
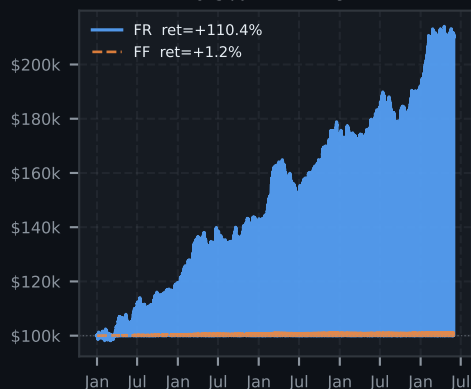
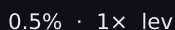
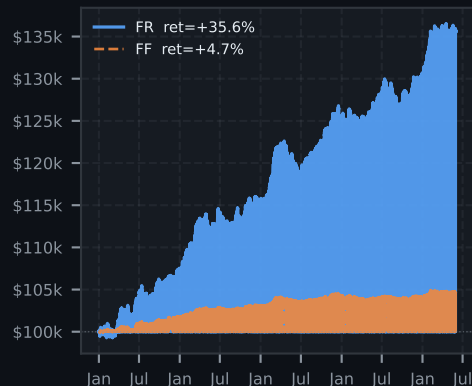
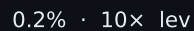
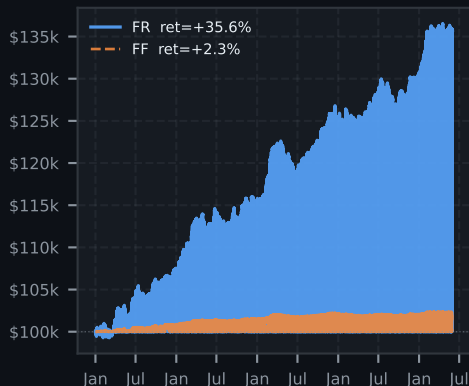
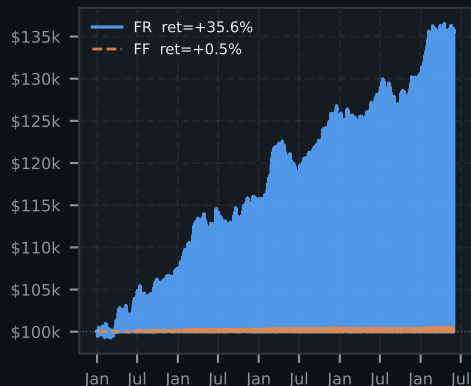
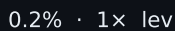
H · Monte Carlo - Percentile Summary Table

Scenario	Median Ret (%)	p5 Ret (%)	p95 Ret (%)	Median MDD (%)	Worst MDD 5% (%)	Median Sharpe	P(Profit) (%)	P(Ruin) (%)
Baseline	512.33	181.47	1153.52	-19.56	-30.87	4.15	100.00	0.00
Session 08-21	273.96	80.00	730.06	-22.33	-34.74	3.12	99.80	0.00
Regime filter	357.73	118.10	855.94	-19.23	-30.11	3.73	99.80	0.00



I · Leverage & Sizing - Equity Curves [Session 08-21]

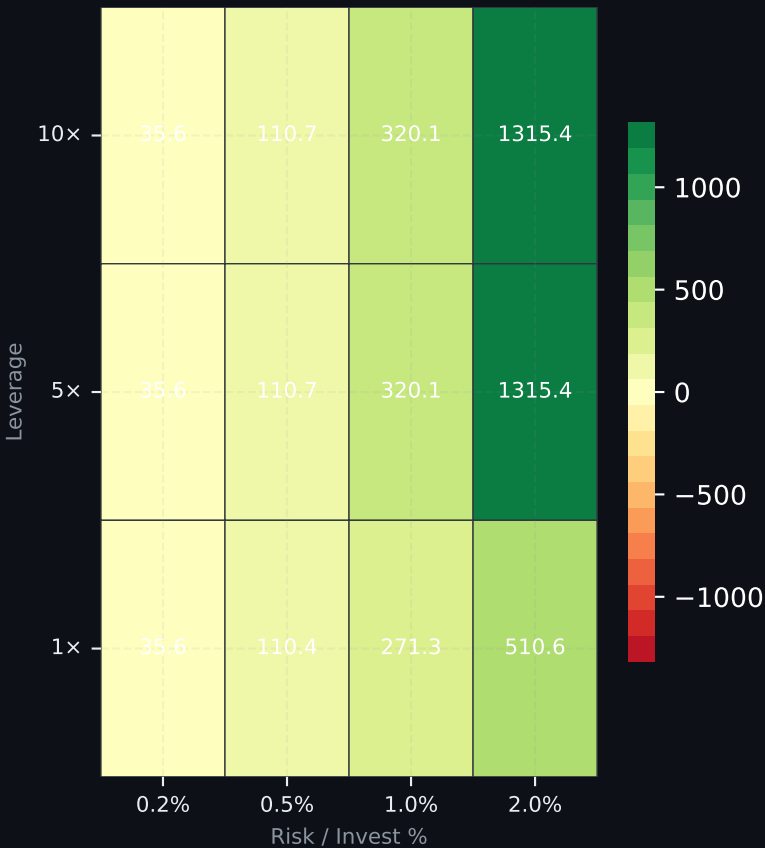
Blue = Fixed Risk · Orange = Fixed Fraction · Rows = % level · Cols = leverage



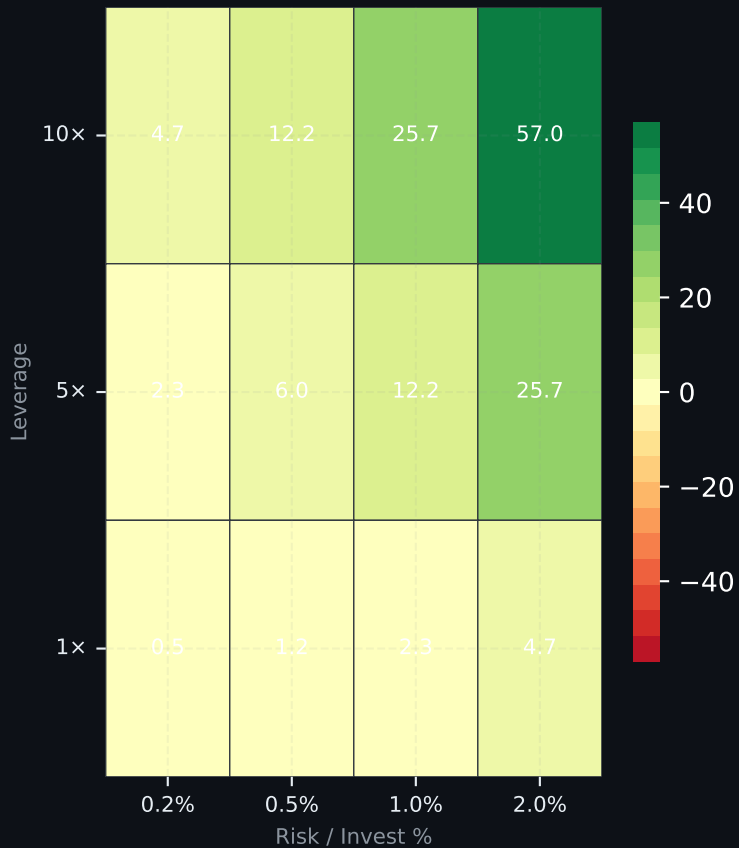
J · KPI Heatmaps - Leverage × Risk Level [Session 08-21]

Rows = Leverage · Cols = % level · Left = Fixed Risk · Right = Fixed Fraction

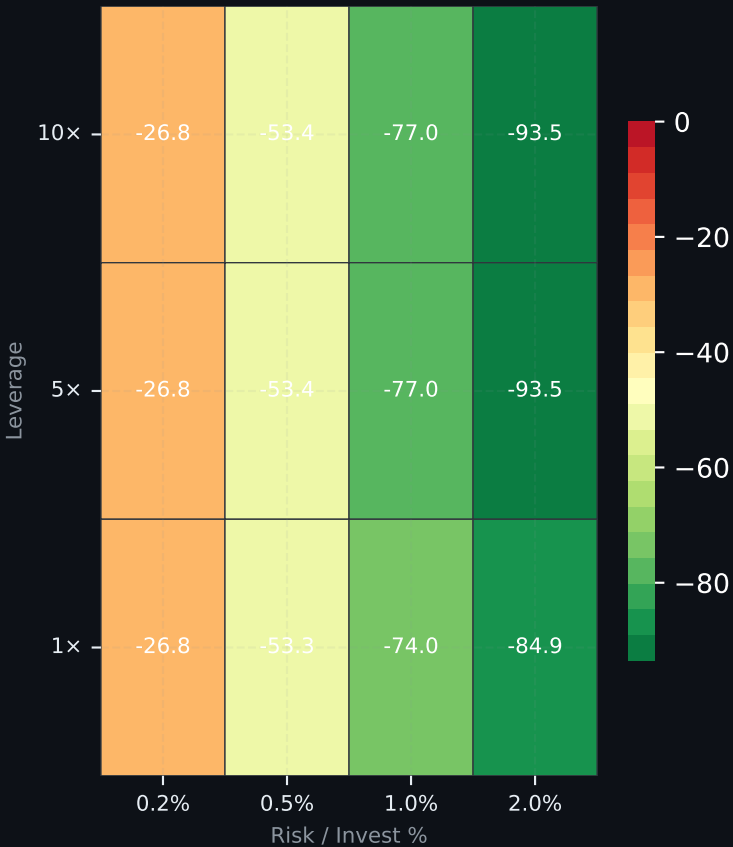
Fixed Risk - Total Ret (%)



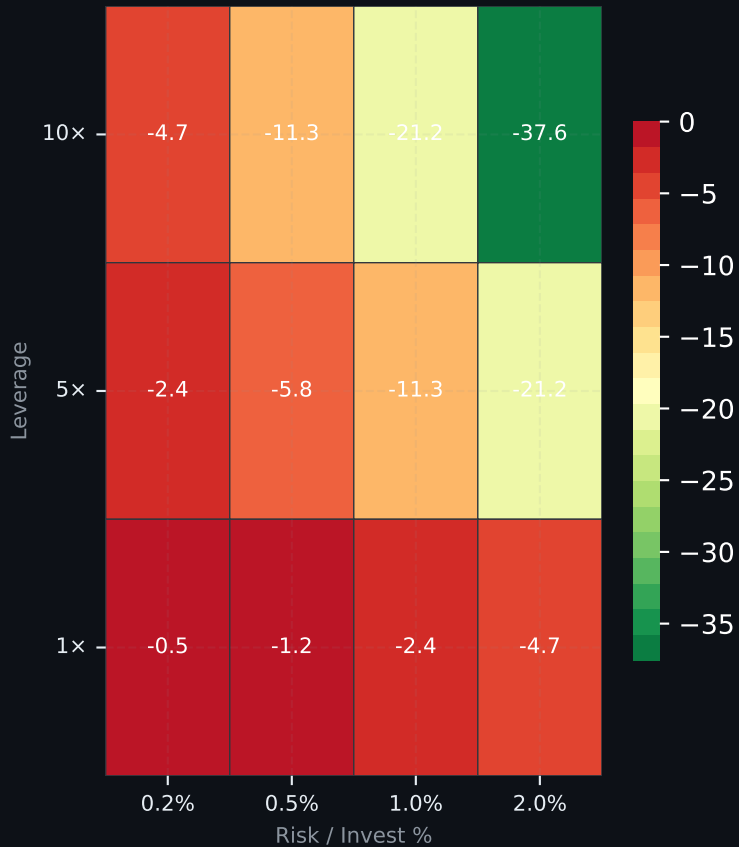
Fixed Fraction - Total Ret (%)



Fixed Risk - Max DD (%)



Fixed Fraction - Max DD (%)

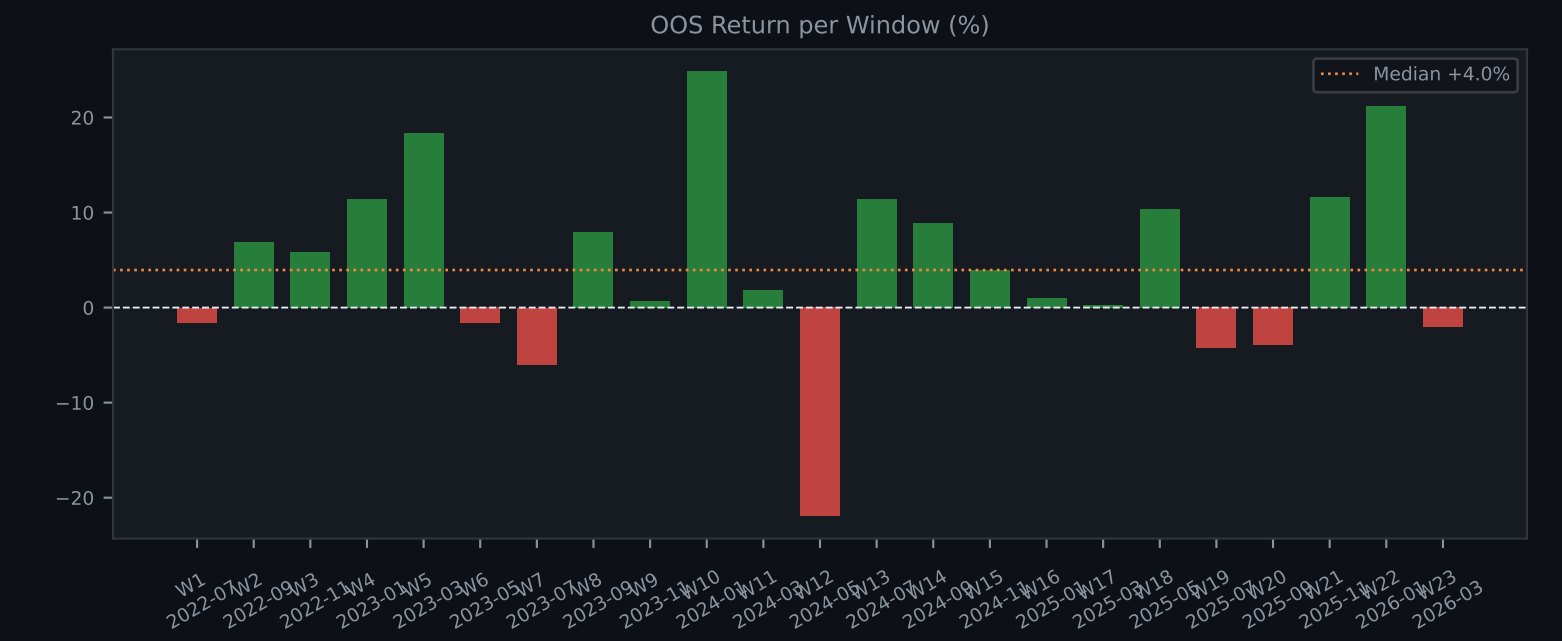
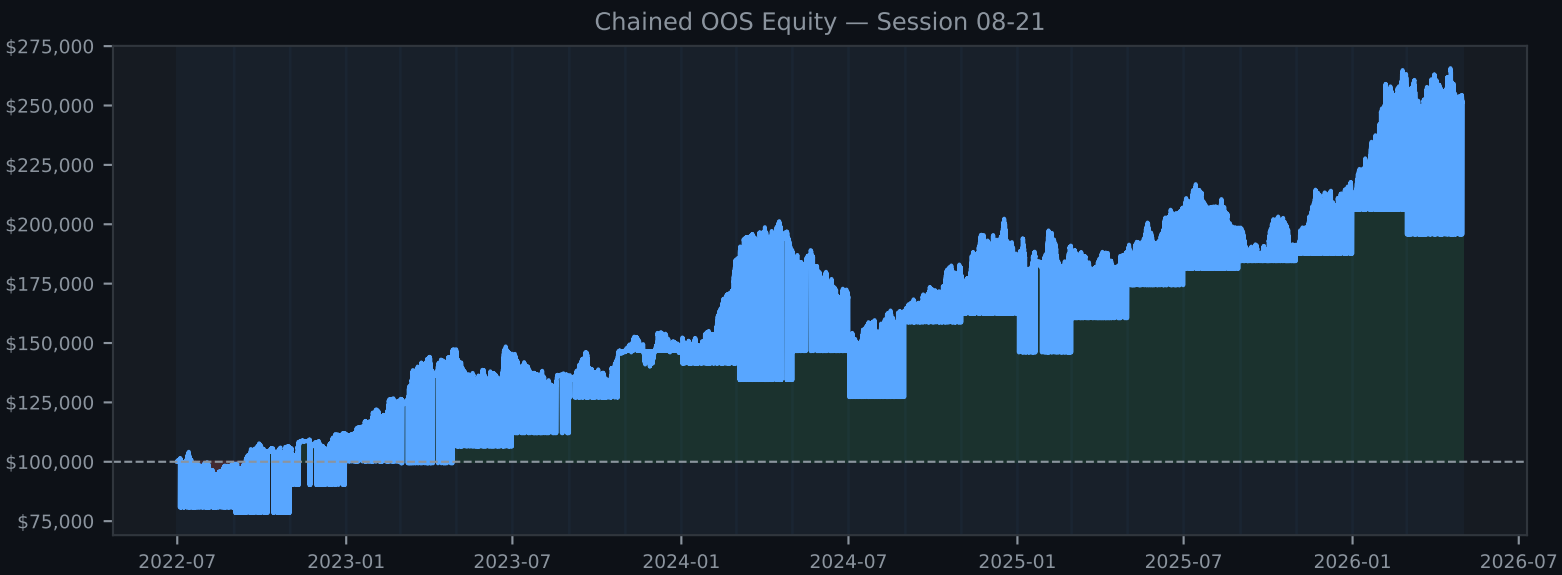


K · Best Configurations Ranking (by Sharpe Ratio)

Top 15 across all scenarios × sizing methods × leverage × risk levels



L · Walk-Forward Validation (OOS)



23

Windows tested

70%

Profitable windows

+4.0%

Median OOS return

2.773

OOS Sharpe (combined)

-36.8%

OOS Max DD (combined)

+0.464

Consistency (Sharpe of rets)

M · Walk-Forward Statistics

